

Middleware Architectures 2

Lecture 7: Kubernetes Storage

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Overview

- Storage Fundamentals
 - *Introduction to Storage in Kubernetes*
 - *Core Storage Abstractions*
- Storage Provisioning
- Storage Backends

Why Persistent Storage Matters

- Containers are **ephemeral** by design
 - *When a container stops or restarts, its filesystem is discarded*
 - *Data written inside a container is lost on Pod deletion or reschedule*
- Many real-world applications require persistent data
 - *Databases (MySQL, PostgreSQL, MongoDB)*
 - *Message queues (Kafka, RabbitMQ)*
 - *File uploads, caches, configuration stores*
- Kubernetes must provide a way to decouple data lifecycle from Pod lifecycle
 - *Data must outlive the Pod that created it*
 - *Storage must be accessible when a Pod is rescheduled to another node*
 - *Multiple Pods may need to share the same data*

Storage Challenges

- Data persistence
 - *Default container storage is tied to the container runtime layer (OverlayFS)*
 - *All writes are lost when a container is removed*
- Portability across nodes
 - *Pods can be rescheduled to any node in the cluster*
 - *Node-local storage is not available after rescheduling*
- Multi-tenancy and isolation
 - *Different applications must not access each other's data*
 - *Storage should be namespaced and access-controlled*
- Diverse storage backends
 - *Teams use different storage solutions: NFS, iSCSI, cloud block/file storage*
 - *Kubernetes must abstract over all of them through a common API*
- Shared access
 - *Some workloads need multiple Pods to read/write the same volume simultaneously*
 - *Access semantics differ across storage backends (block vs. file vs. object)*

Kubernetes Storage Model Overview

- Kubernetes provides a layered abstraction for storage
 - *Applications request storage without knowing the underlying backend*
 - *Administrators configure and expose storage resources*
- Key objects in the storage model
 - **Volume** – *ephemeral or persistent storage attached to a Pod*
 - **PersistentVolume (PV)** – *a storage resource in the cluster*
 - **PersistentVolumeClaim (PVC)** – *a request for storage*
 - **StorageClass** – *a template for dynamically provisioning PVs*
 - **CSI Driver** – *plugin interface to external storage systems*
- Two provisioning models
 - **Static** – *administrator creates PVs manually in advance*
 - **Dynamic** – *Kubernetes creates PVs automatically on demand*

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Volumes

- A **Volume** is a directory accessible to containers in a Pod
 - *Declared in the Pod spec; mounted into one or more containers*
 - *Lifecycle is tied to the Pod (not the individual container)*
 - *Data in a volume survives container restarts within the same Pod*
- Common ephemeral volume types
 - **emptyDir** – *scratch space created when Pod starts; deleted with Pod*
 - **configMap** / **secret** – *inject configuration or credentials as files*
 - **downwardAPI** – *expose Pod metadata to containers as files*
- Usage example

```
1 spec:
2   volumes:
3     - name: cache-vol
4       emptyDir: {}
5   containers:
6     - name: app
7       image: myapp:1.0
8       volumeMounts:
9         - mountPath: /cache
10          name: cache-vol
```

PersistentVolume (PV)

- A **PersistentVolume** is a cluster-level storage resource
 - *Provisioned by an administrator or dynamically by a StorageClass*
 - *Independent lifecycle from any individual Pod*
 - *Backed by a real storage system (NFS, cloud disk, local path, etc.)*
- Key PV properties
 - **Capacity** – storage size (e.g. **10Gi**)
 - **Access Modes** – how the volume can be mounted (see next slide)
 - **Reclaim Policy** – what happens when the PV is released
 - **Volume Mode** – **Filesystem** (default) or **Block**
 - **StorageClass** – class used for binding and provisioning
- PV phases
 - **Available** → **Bound** → **Released** → **Failed**

PV Access Modes and Reclaim Policies

- Access Modes
 - **ReadWriteOnce (RWO)** – *mounted read-write by a single node*
 - **ReadOnlyMany (ROX)** – *mounted read-only by nodes*
 - **ReadWriteMany (RWX)** – *mounted read-write by nodes simultaneously*
 - **ReadWriteOncePod (RWOP)** – *read-write by a single Pod*
- Not all backends support all access modes
 - *Block storage (AWS EBS, GCE PD) typically supports only **RWO***
 - *Shared filesystems (NFS, Azure Files) support **RWX***
- Reclaim Policies
 - **Retain** – *PV is kept after PVC deletion; manual cleanup required*
 - **Delete** – *PV and underlying storage are deleted with the PVC*

PersistentVolumeClaim (PVC)

- A **PersistentVolumeClaim** is a request for storage by a user
 - *Declares required capacity, access mode, and a StorageClass*
 - *Kubernetes binds the PVC to a matching PV*
 - *The bound PV is then mountable in a Pod*
- PVC is namespace-scoped; PV is cluster-scoped
- Example PVC and Pod using it

```
1  apiVersion: v1
2  kind: PersistentVolumeClaim
3  metadata:
4    name: my-pvc
5  spec:
6    accessModes:
7      - ReadWriteOnce
8    resources:
9      requests:
10       storage: 5Gi
11    storageClassName: standard
```

```
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: my-app
5  spec:
6    volumes:
7      - name: data
8        persistentVolumeClaim:
9          claimName: my-pvc
10    containers:
```

StorageClass

- A **StorageClass** defines a template for dynamic PV provisioning
 - *Specifies the provisioner (CSI driver or in-tree plugin)*
 - *Carries parameters specific to the storage backend*
 - *Defines the default reclaim policy for dynamically created PVs*
- Benefits
 - *Different classes can represent different tiers: fast SSD vs. slow HDD*
 - *Developers request storage by class name without knowing the backend*
 - *One cluster can offer multiple classes for different use cases*
- Example

```
1  apiVersion: storage.k8s.io/v1
2  kind: StorageClass
3  metadata:
4    name: fast-ssd
5  provisioner: ebs.csi.aws.com
6  parameters:
7    type: gp3
8    encrypted: "true"
9  reclaimPolicy: Delete
10 volumeBindingMode: WaitForFirstConsumer
```

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 - *Static Provisioning*
 - *Dynamic Provisioning*
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What is Static Provisioning?

- Administrator creates Persistent Volumes **manually** in advance
 - *The underlying storage resource must already exist*
 - *PV spec describes how to connect to the resource (NFS share, disk path, etc.)*
- Workflow
 1. *Admin provisions storage resource (e.g. creates NFS export)*
 2. *Admin creates a PV object pointing to that resource*
 3. *Developer creates a PVC requesting matching properties*
 4. *Kubernetes binds the PVC to the existing PV*
 5. *Pod mounts the volume via the PVC*
- Good for
 - *Small clusters or on-premise setups with fixed storage infrastructure*
 - *Workloads with specific, pre-configured storage requirements*
 - *Environments where an administrator controls every storage resource*

Static Provisioning Example

- Admin creates a PV backed by an NFS share

```
1  apiVersion: v1
2  kind: PersistentVolume
3  metadata:
4    name: nfs-pv
5  spec:
6    capacity:
7      storage: 20Gi
8    accessModes:
9      - ReadWriteMany
10   persistentVolumeReclaimPolicy: Retain
11   nfs:
12     server: 192.168.1.100
13     path: /exports/data
```

- Developer creates a matching PVC

```
1  apiVersion: v1
2  kind: PersistentVolumeClaim
3  metadata:
4    name: nfs-pvc
5  spec:
6    accessModes:
7      - ReadWriteMany
8    resources:
9      requests:
10       storage: 20Gi
```

- Kubernetes finds the PV that satisfies the PVC criteria and binds them

Static Provisioning – Pros and Cons

- Pros
 - *Full administrator control over every storage resource*
 - *Works with any storage backend regardless of CSI support*
 - *Predictable: storage is prepared before workloads request it*
 - *Regulated environments requiring manual approval workflows*
- Cons
 - *Requires manual effort for every new storage resource*
 - *Does not scale well in large or dynamic clusters*
 - *Risk of PV/PVC mismatch causing pending claims*
 - *Capacity may be under-utilized if not carefully matched*
 - *No automation – developers must wait for admin action*

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What is Dynamic Provisioning?

- Kubernetes automatically creates a PV when a PVC is submitted
 - *PVC references a **StorageClass** instead of a specific PV*
 - *The StorageClass provisioner calls the storage backend API*
 - *The volume is allocated*
 - *The PV is created, bound to the PVC, and ready for use*
- Workflow
 1. *Administrator deploys a StorageClass*
 - *Defines the provisioner (e.g. CSI driver), parameters for the backend*
 2. *Developer creates a PVC referencing the StorageClass*
 3. *Kubernetes provisioner creates a new PV on the backend*
 4. *PVC is bound to the new PV; Pod can mount it immediately*
- Supported by all major cloud providers and on-premise solutions
 - *AWS EBS, GCE PD, Azure Disk, OpenEBS, etc.*

Dynamic Provisioning Example

- StorageClass for AWS EBS using the CSI driver

```
1  apiVersion: storage.k8s.io/v1
2  kind: StorageClass
3  metadata:
4    name: aws-ebs-sc
5    annotations:
6      storageclass.kubernetes.io/is-default-class: "true"
7  provisioner: ebs.csi.aws.com
8  parameters:
9    type: gp3
10 reclaimPolicy: Delete
11 volumeBindingMode: WaitForFirstConsumer
```

- Developer creates a PVC (PV does not exist)

```
1  apiVersion: v1
2  kind: PersistentVolumeClaim
3  metadata:
4    name: app-data
5  spec:
6    accessModes:
7      - ReadWriteOnce
8    storageClassName: aws-ebs-sc
9    resources:
10     requests:
11       storage: 10Gi
```

- Kubernetes calls the EBS CSI driver
 - *new EBS volume is created and bound*

Dynamic Provisioning – Pros and Cons

- Pros
 - *Self-service for developers – no admin intervention required*
 - *Scales automatically with cluster demand*
 - *Eliminates idle pre-provisioned storage*
 - *Integrates with cloud cost management (pay-per-use)*
 - *Supports version control of storage configuration via StorageClass*
- Cons
 - *Requires a working CSI driver or in-tree provisioner*
 - *Less control over individual storage resources*
 - *Uncontrolled PVC creation can lead to unexpected cloud costs*
 - *Debugging provisioning failures can be complex (CSI logs, events)*
 - *Not always available in air-gapped or strictly controlled environments*

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- Storage Backends
 - *Local Node Storage*
 - *External Storage*

hostPath Volumes

- Mounts a file/directory from the **node's filesystem** into the Pod
 - *Directly exposes a node path such as **/var/log** or **/data***
 - *The container reads and writes directly to the node's disk*
- Common use cases
 - *Log collection agents that need access to **/var/log***
 - *Monitoring tools accessing **/proc** or **/sys***
 - *Development and testing on single-node clusters*
- Risks and limitations
 - *Tightly couples the Pod to a specific node*
 - *If the Pod is rescheduled to another node, the data is not available*
 - **Security risk:** *container can read/modify sensitive host files*
 - *Not suitable for multi-node production workloads*
- This should generally be avoided for persistent application data in production

Local Persistent Volumes

- A **local** PersistentVolume uses a dedicated disk or partition on a node
 - Defined with **local** volume type in the PV spec
 - Unlike **hostPath**, local PVs support standard PV/PVC lifecycle
 - Kubernetes topology-aware scheduling ensures Pods land on the correct node
- Example

```
1  apiVersion: v1
2  kind: PersistentVolume
3  metadata:
4    name: local-pv
5  spec:
6    capacity:
7      storage: 100Gi
8    accessModes:
9      - ReadWriteOnce
10   persistentVolumeReclaimPolicy: Retain
11   storageClassName: local-storage
12   local:
13     path: /mnt/disks/ssd1
14   nodeAffinity:
15     required:
16       nodeSelectorTerms:
17         - matchExpressions:
18           - key: kubernetes.io/hostname
19             operator: In
20             values:
21               - worker-node-1
```

Local Node Storage – Pros and Cons

- Pros
 - *Very high I/O performance – direct access to local NVMe or SSD disks*
 - *Low latency; no network overhead*
 - *Simple setup; no external storage system required*
 - *Good for read-heavy workloads, caches, and stateful databases*
- Cons
 - *Data is bound to a specific node – no built-in replication*
 - *Node failure means data loss unless application handles replication itself*
 - *Pod scheduling must respect node affinity – reduces scheduling flexibility*
 - *Scaling out requires adding disks to specific nodes manually*
 - ***hostPath** is insecure and unsuitable for multi-tenant clusters*
- Recommended pattern: use local volumes only with applications that manage their own data replication (e.g. distributed databases)

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 - *Local Node Storage*
 - *External Storage*

Network File System (NFS)

- NFS is a widely used network-attached filesystem
 - *A central NFS server exports directories shared over the network*
 - *Multiple Pods can mount the same NFS share simultaneously (ReadWriteMany)*
 - *Supported natively in Kubernetes via the `nfs` volume type*
- Typical setup
 - *NFS server runs on a dedicated machine or network appliance*
 - *Kubernetes nodes mount NFS exports; data is network-accessible from all nodes*
 - *Dynamic provisioning available via the `nfs-subdir-external-provisioner`*
- Example PV using NFS

```
1 spec:
2   capacity:
3     storage: 50Gi
4   accessModes:
5     - ReadWriteMany
6   nfs:
7     server: nfs.example.com
8     path: /exports/shared
```

- Common for shared content, logs aggregation, and legacy workloads

Cloud Block Storage and CSI

- Cloud providers offer managed block storage backends
 - *AWS: Elastic Block Store (EBS)* – provisioned via `ebs.csi.aws.com`
 - *GCP: Persistent Disk (PD)* – provisioned via `pd.csi.storage.gke.io`
 - *Azure: Managed Disk / Azure Files* – via `disk.csi.azure.com`
- **Container Storage Interface (CSI)**
 - *Standard API between Kubernetes and storage plugins*
 - *Decouples storage drivers from the Kubernetes core*
 - *CSI driver is deployed as a DaemonSet/Deployment in the cluster*
 - *Exposes capabilities: provision, attach, mount, snapshot, resize, clone*
- CSI driver operations
 - `CreateVolume` / `DeleteVolume` – *lifecycle management*
 - `NodePublishVolume` / `NodeUnpublishVolume` – *mount/unmount on node*
 - `CreateSnapshot` – *point-in-time backup*
- On-premise options: Ceph, iSCSI, OpenEBS

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 - *Advanced Topics and Best Practices*

StatefulSet with Persistent Storage

- **StatefulSet** is the workload resource designed for stateful applications
 - *Each Pod gets a stable identity (name, DNS) and its own PVC*
 - *PVCs are created from a **volumeClaimTemplates** section*
 - *PVCs survive Pod deletion and are reattached when the Pod is recreated*
- Example

```
1  apiVersion: apps/v1
2  kind: StatefulSet
3  metadata:
4    name: mysql
5  spec:
6    serviceName: mysql
7    replicas: 3
8    template:
9      spec:
10     containers:
11       - name: mysql
12         image: mysql:8.0
13         volumeMounts:
14           - name: data
15             mountPath: /var/lib/mysql
16     volumeClaimTemplates:
17       - metadata:
18         name: data
19       spec:
20         accessModes: [ "ReadWriteOnce" ]
21         storageClassName: fast-ssd
22         resources:
23           requests:
24             storage: 20Gi
```

- Each Pod (e.g. **mysql-0**, **mysql-1**) gets its own **data-mysql-N** PVC

Volume Snapshots

- Kubernetes supports point-in-time snapshots of PersistentVolumes
 - *Requires a CSI driver that implements snapshot capabilities*
 - *Snapshots are managed via **VolumeSnapshot**, **VolumeSnapshotContent**, and **VolumeSnapshotClass** resources*
- Use cases
 - *Backup and disaster recovery*
 - *Creating test environments from production data*
 - *Pre-upgrade checkpoints*
- Creating a snapshot

```
1  apiVersion: snapshot.storage.k8s.io/v1
2  kind: VolumeSnapshot
3  metadata:
4    name: db-snapshot
5  spec:
6    volumeSnapshotClassName: csi-aws-vsc
7    source:
8      persistentVolumeClaimName: db-pvc
```

Storage Best Practices

- Choose the right storage type for the workload
 - *Use local volumes only for applications that replicate data themselves*
 - *Use shared (RWX) storage only when multiple Pods need concurrent access*
 - *Default to cloud block storage (CSI, RWO) for most stateful workloads in cloud*
- Use dynamic provisioning wherever possible
 - *Reduces operational overhead and scales automatically*
 - *Define StorageClasses for different performance tiers*
- Set appropriate reclaim policies
 - *Use **Retain** for production data to prevent accidental deletion*
 - *Use **Delete** for ephemeral or dev/test environments*
- Enable volume snapshots for critical workloads
- Apply resource quotas on PVC requests per namespace to control costs
- Monitor storage utilization; avoid PV over-provisioning